

Sprint 3 Intermediate Product Report

Team Name: AI-Driven Robotic Manipulation: Integrating Learning Models with Real-World Control

Team Members: Toan Pham, Albert Vo

Sponsor: Dr. Nakul Gopalan

Date: Oct 21, 2025

1. Overview

Sprint Goal: Achieve initial cross-embodiment transfer between Franka and Sawyer robots

Work Completed:

SBI-301 – Sawyer Robot Configuration (In Progress)

2. Achievements

User Stories:

Cross-Embodiment Foundation: Sawyer Robot Configuration – In Progress

3. Risk Management

Critical Risk Materialized

Risk F: Kinematic Mismatch

Status: Occurred

Impact: High - Sawyer cannot reach some Franka positions

Resolution: Implemented workspace scaling and task redesign

Lessons: Should have analyzed workspace overlap earlier

Owner: Albert Vo

New Risks

Risk G: Transfer Performance Gap

Status: Monitoring

Probability: High

Impact: High - Core project goal

Mitigation: Multiple transfer strategies prepared

Owner: Both

Risk H: Training Instability

Status: Active

Probability: Medium

Impact: Medium - Delays convergence
Mitigation: Gradient clipping, learning rate scheduling
Owner: Toan Pham

4. Contribution Metrics

Velocity: 31 story points (of 40 planned)
Task Completion: 28 of 35 tasks (80%)
GPU Hours Used: 187 hours
Total Demonstrations: 9 (Franka: 5, Sawyer: 4)
Cross-Embodiment Success:
Franka→Sawyer: 38%
Sawyer→Franka: 41%
Within-robot baseline: 52%

5. Evidence

Taiga Board: https://tree.taiga.io/project/chickeninvader-chickeninvadercross_model_learning_based_robot_control/us/6?kanban-status=10491383
GitHub Repository: <https://github.com/ChefToan/Cross-Model-Learning-Robot-Control.git>

6. Feedback Summary

Dr. Gopalan's Assessment

"This sprint demonstrates genuine research progress. The 38-41% transfer rate is submission-ready, especially with scene graph mediation. For Sprint 4, focus on understanding why certain tasks transfer better and consider curriculum learning approaches."

Recommendations

7. Updates

Adjustments: No major changes beyond normal sprint refinement.
New or Withdrawn EPICs/User Stories: Updated on the product backlog as needed.
Changes in Priority: Priorities adjusted based on sprint review outcomes.

8. Signature from the Product Owner/Client

Signature/email:

A handwritten signature in red ink, consisting of a large, stylized 'X' or '10' shape followed by a horizontal line.